

Fundamentals of Data Management (FSPH B452/B552)
FINAL PROJECT
Due: May 2, 2025

Given the study protocol, the five SAS data sets, the data dictionaries and the scoring document found in the Final Project assignment tab on Canvas, follow the guidelines for **Preparing data for statistical analyses** outlined in Lecture 11 Slides 4-22 to achieve the following:

1. Clean the data in SAS
 - a. Identify and print the duplicate records. Based on the records in other data sets, make educated guesses as to which of the duplicate records is likely correct and eliminate or modify the others. This guess should be based off the visit dates of the records. Document your actions in the dataset cover sheet (see #4 below).
 - b. In the CATIS and Family Resources data files, identify and print records with values that are out of range such as RES1=5 in the FAMRES_FP dataset. Determine how best to deal with these values. Document your decisions in the dataset cover sheet (see #4 below).
 - c. For the Family Resources questionnaire, ignoring visit number:
 - i. calculate the number of days between visits
 - ii. generate frequencies grouping the days into 3 categories: < 1 year, 1-2 years and >=2 years
 - iii. generate a clean-up query list, displaying all FAMRES records, for participants who have less than one year or more than 2 years between visits. Provide a meaningful title for this list.
2. Generate a permanent cross-sectional analysis data set.
 - a. Using PUT, SCAN and LEFT functions, extract the state from the formatted site code. Hint: to get started, see Program 11.14 in textbook.
 - b. Label and format (as needed) all variables.
 - c. Generate frequencies for demographic data where appropriate.
 - d. Run PROC Contents on this analysis data set.
3. Generate permanent longitudinal analysis data set(s).
 - a. Calculate child's age at each encounter.
 - b. Generate summary scores for the data collection instruments CATIS, Family Resources, and Infant Characteristics Questionnaire (ICQ) based on the criteria in score_notesSZ_FP.doc.
 - c. Label and format (if needed) all derived variables such as summary scores. *You do not need to label or format the individual questionnaire variables such as RES1-RES30.*
 - d. Run PROC Contents on this/these analysis data set(s).
4. Create a dataset cover sheet describing your analysis datasets.
5. For each of the summary score variables, generate the following statistics: number of missing values, number of nonmissing values, mean, standard deviation, minimum, first quartile (Q1), median, third quartile (Q3) and maximum. Report these statistics overall and broken down by sex.

6. Produce a horizontal bar chart showing a frequency distribution of race. Within each bar, show the distribution of sex.
7. Submit your SAS program file, log, and output along with your permanent SAS data sets, dataset cover sheet.